Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 1 AVSummary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 1 Class 1 AVSummary

1 UE26ME142A Unit 1 Class 1 AVSummary

Study focus: UE26ME142A_Unit 1_Class 1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000018 · EV0000019

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Dear student, welcome all of you for engineering mechanics classes.

Key details

- I am Dr.
- Svinkitesh, civil engineering from the Department of Science and Humanities.
- I welcome all of you for the first year courses under PS University.
- I will be teaching you the subject engineering mechanics statics.
- which is a common subject for all students who are entering into B Tech program under PS University.
- I will start with the first chapter, which is Introduction to Statics.
- For this subject, we are using a textbook by the title Engineering Mechanics Statics.
- And we are using the eighth edition.
- This addition is in SI version.
- for this book is JL Meriam and LG Craig.
- And you can see here the front covering page or the face page of the book.
- It comes under two different colours, but the contents are same.
- The eighth edition is adopted by.
- J and Bolton and that's what you can see here.
- The author's name is LJ Craig.
- Sorry, J L Miriam, L G Craig, J and Bolton, and it's the same thing here also.
- But only the difference between these two is the colour.
- One is in red colour, the other one is in green colour.
- You can buy whichever is available in the market.

- Both of them, the content inside is same, so you can take up whichever book that is available in the market.
- So in the first introduction chapter, we will be dealing with this.
- First thing is what is mechanics?
- Mechanics is that branch of engineering which deals with the study of bodies which are under rest.
- Which are in motion.
- There are two aspects for this.
- The first one is the bodies which are at rest, which is called as statics.
- The bodies which are in motion.
- comes under dynamics.
- One is the steady of bodies which are at rest which is called statics.
- The other one is the bodies which are in motion which is comes under dynamics.
- Few of the basic definition that we need to know is listed here.
- Of course, all this you would have gone through in your earlier classes, which is before joining your MTEC, you would have gone through this.
- And all of you have studied in English medium, or you would have changed from some other medium to English medium.
- Then of course physics.
- This subject is actually an extension of whatever you have learned in your physics.
- There, you would have learned the basics.
- Here you are going to learn the application of the basics with respect to engineering.
- Then you should know mathematics.
- In mathematics, all the basics that you have learned is what is required.
- Then algebra, linear algebra.
- Then trigonometry, then integration, which is calculus.
- And of course very importantly geometry.
- The geometry plays a very important role when you have to solve some problems.
- As we go on, you will come to know what are the things that we should know.

Example or application from the lecture

- For example, arithmetic, which is addition, subtraction, multiplication, and division must ensure.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 05:00–09:59

All of you would have known what is space.

Key details

- How do you define space?
- You should know what is space.
- Space is the object.

- which occupies the point which can be measured with respect to the three coordinate systems.
- We have three coordinate systems for defining any point on this earth or in surrounding.
- We need to, we or we ref we refer to three coordinate systems that is x-axis, y axis, and the z axis.
- To know this, we use the right hand first three fingers, that is the thumb, the four finger, and the middle finger, and this are mutually perpendicular to each other.
- If you take the thumb, the thumb if you take it as the x-axis, the forefinger represents the y axis and the middle finger represents the z axis.
- By the three coordinate systems, the distance as X, Y, and Z, which talks about what is the X distance of that point with from the origin measure along the x-axis.
- Then y distance is what is the distance measured from the origin.
- Along the y axis, then Z distance is the distance measured from the origin along the z axis.
- This is one way of identifying a point in space with respect to the three different axes.
- We can also locate.
- The point by using the angular measurements.
- That is what is the angle made by this point in space with respect to the x axis that is theta x, then with respect to y axis that is theta y, then with respect to the z axis that is theta z.
- Either we can have this three distance measurement or three angular measurement or the combination of this distance measurement or the angular measurement.
- Between this is 3D.
- 3D is when we use all the three coordinate systems, we go for what is called as three-dimensional objects.
- Then we can define that or we can represent it as a two-dimensional object.
- 3D is when we have to define.
- The object by all the three coordinate systems, then we go for 3D, or it is called as three dimensional objects.
- The next in the definition is mass.
- What is mass?
- It is the amount of substance that is present in a body is called as mass.
- Then how do we define time?
- Time is defined as the measure of succession of events.
- The instant that we leave our house and the instant that we reach the college, that measure between the instant that we leave our house and the instant that we reach the college, that measure of succession of events is called as time.
- Next, how do we define a particle?

Example or application from the lecture

- If we can represent the object and the forces acting on them along any one plane using only two coordinate systems, for example, only X and Y.
- For example, if we have to measure the or we have to know the time elapsed between.
- We don't consider time when we are talking about objects which are at rest, that is, when we are dealing with statics.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 10:00–14:59

A particle is defined as an object which occupies a single point in space which has mass, which has negligible dimension.

Key details

- With respect to naked eyes.
- Of course, when you go to nano level, you can certainly measure the dimensions of a particle.
- Point which is occupied by the object in space, one single point.
- That means we can measure it with respect to the coordinate system X, Y, and Z.
- Then when we say it has mass, the particle will settle down.
- Then it is very tiny and we cannot measure it with the naked eyes.
- Such objects are defined as particles.
- Next, the three different types of bodies.
- What is a rigid body?
- A rigid body is one in which when we apply force onto this, there is no change in shape and size of the body.
- Such bodies are defined as rigid body.
- Next, elastic body.
- What is an elastic body?
- When a force is applied on an elastic body, it changes its shape and size.
- But on removal of the force causing this change, the body will regain its original shape and size.
- Such bodies are called as elastic bodies.
- Next is a plastic body.
- What is a plastic body?
- A plastic body is one.
- Which will change its shape and size on application of a force, but it will not regain its original shape and size.
- Such bodies are called as plastic bodies.
- Next, how do we define a force?
- A force is something which changes the state of rest of uniform motion of an object.
- If the object or the body is already in motion, it can stop that motion or it can increase the speed of the body or it can reduce the speed of the body.
- For such application or such thing is called as a force.
- Now, next comes what are the different types of forces.
- Forces can be broadly divided into two types.
- Forces which are having forces which are in one plane, coplanar forces.
- The other one is non-coplanar forces.
- Which can be represented only along two dimensions.
- Non-co-planar forces are forces which can be represented along all the three coordinate systems that is required.

- Concurrent forces and non-concurrent forces.
- What are concurrent forces?
- Concurrent forces are forces which will have the same point of application.
- Non-concurrent forces which will have different points of application.
- Under non-concurrent force system, we can have parallel force system and non-parallel force system.
- Under parallel force system, all the forces are going to be parallel to each other.
- Non-parallel force system, they need not have to be parallel.
- Under parallel force system, we can have two categories.
- One is like parallel force system and unlike parallel force system.
- What is like parallel force system?
- Like parallel force system, where the sense of the force are same for all.
- Unlike parallel force system, the sense of the forces are different.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next coming to what is the you what are the units of forces.

Key details

- Unit of forces in SI, that is system of interaction.
- SI is Newtons.
- And you can see the picture of Sir Isaac Newton.
- Which is pulled by the gravitational force.
- 9.81 newtons means it is one kg of force which is pulled by a gravitational force that is 9.81.
- 1 into 9.81 meters square meter per second square.
- What are Newton's laws?
- There are three Newton laws.
- All of you have studied this.
- Newton's law one, two, and three.
- What is Newton's law one?
- Newton Law 1 states that.
- Any body would continue to be in its state of rest or of uniform motion unless and until an external force is applied onto it.
- Is what you have studied.
- But here, instead of saying external force, we say that unless and until it is.
- It is unbalanced force that is acting.
- Where is that force coming?
- It is because the internal that is developed by the person.
- A body will continue to be in its state of rest or of uniform motion unless and until there is an unbalanced force acting on it.
- The second law states that the force is proportional to the mass of the body.
- Acceleration is the proportionality constant.

- F is equal to M.
- Of course, the Newton's third law, all of every action has an equal and opposite reaction.
- Which you already know or studied.
- That is what are the characteristics of forces.
- What is the characteristics of force?
- Force one is the magnitude.
- Magnitude is defined as the amount of force that is acting.
- The amount of quantity of the force is called as magnitude.
- Next is the direction.
- The direction is the angle made by the force with respect to a reference axis.
- The reference axis can be anything.
- The third one is the.
- Point of application.
- Where are we supposed to apply this?
- That is the point of application.
- The fourth one is the sense of the force.
- Whether it is acting away from the point or towards the point.
- In the book, it clubs the direction and the sense together.
- Garney says that line of action of the force.
- Then your direction if it is theta and if it is moving towards the angle will become 180 plus theta.
- So this much we can understand in the first class.
- We will continue this in the next class.
- And I am Dr.
- Svang Tesh, civil engineering.
- from the Department of Science and Humanities, PS University.
- See you in my next class where we will be discussing further into the introduction of statics.

Example or application from the lecture

- For example, a person who is standing, he can walk.
- Let's look at this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.

4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare...
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare...

C Revision and retrieval practice